Tommaso Bonato

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EDUCATION

ETH Zürich | Ph.D. in Computer Science Jan 2023 – Mar 2027 (expected)

- Part of the Scalable Parallel Computing Lab (SPCL), advised by Prof. Torsten Hoefler.
- Research: AI workloads, collective communication, congestion control, and load balancing for AI/HPC.
- Publications in venues including SC, NSDI, SIGCOMM, EuroSys, and *Communications of the ACM*.
- Supervised and mentored several bachelor's and master's students, guiding research design and execution.
- Teaching Assistant: Wireless Networking and Mobile Computing, Fall 2026.

ETH Zürich | M.S. in Computer Science Sep 2018 – Nov 2021

- GPA: 5.31/6.00. Thesis on adaptive routing in the Slingshot interconnect, with Hewlett-Packard.

University of Bologna | B.S. in Computer Science Sep 2014 – Mar 2018

- Graduated with honors, 110/110; GPA: 29.1/30.
- Scholarship-funded exchange at the University of California, Santa Cruz (2016–2017).

WORK EXPERIENCE

NVIDIA | HPC and AI Software Network Architect Intern Jun 2026 – Nov 2026

- Design and evaluate multi-NIC load balancing for GPU-initiated networking (GIN) in AI systems.
- Contributed to the **production release** of a multi-NIC load-balancing feature for NVIDIA Vera Rubin.

Microsoft (via Experis) | Research Engineer (Part-Time) Aug 2022 – Mar 2026

- Designed transport protocols for Azure's AI/LLM and HPC infrastructure in the Network Architecture team.
- Represented Microsoft in the Ultra Ethernet (UE) Transport Working Group, contributing to congestion control and load balancing.
- Developed proofs of concept for inter-datacenter congestion control and reliability, networking/AI co-design, and packet- and flow-level load balancing.
- Extended Microsoft's C++ htsim simulator for **workload-driven evaluation** of AI/HPC networking.

Amazon | Software Engineer Intern Feb 2019 – Jul 2019

- Built an LSTM-based transportation capacity recommendation system using Keras and AWS, contributing to estimated annual savings of **USD 5 million**.
- Built alerts that caught high-priority transportation capacity misconfigurations for Prime Day and holidays.

European Space Agency | Software Engineer Intern Jul 2017 – Oct 2017

- Developed data analysis tools for the James Webb Space Telescope; integrated Sandia National Laboratories' C++ libraries into Fortran code, achieving a **4× speedup**.

SELECTED PUBLICATIONS

- **REPS: Recycled Entropy Packet Spraying for Adaptive Load Balancing and Failure Mitigation.** *EuroSys 2026; first author.* Reference load-balancing design for **Ultra Ethernet (UE)** and inspiration for **Multipath Reliable Connection (MRC)**.
- **ATLAHS: An Application-centric Network Simulator Toolchain for AI, HPC, and Distributed Storage.** *SC 2025; co-first author.* **Best Student Paper Finalist.**
- **Uno: A One-Stop Solution for Inter- and Intra-Data Center Congestion Control and Reliable Connectivity.** *SC 2025; co-first author.*
- **PICO: Performance Insights for Collective Operations.** *ISC 2026.* **Hans Meuer Award (Best Paper).** MPI/NCCL benchmarking and tuning for LLM training.

TECHNICAL SKILLS

- **Proficient:** C/C++, Python. **Familiar:** Java, MPI, NCCL, RDMA, Bash, Git, Docker, Linux.
- **Expertise:** LLM training/inference workloads, GPU communication. **Simulation:** htsim, ns-3, SST.